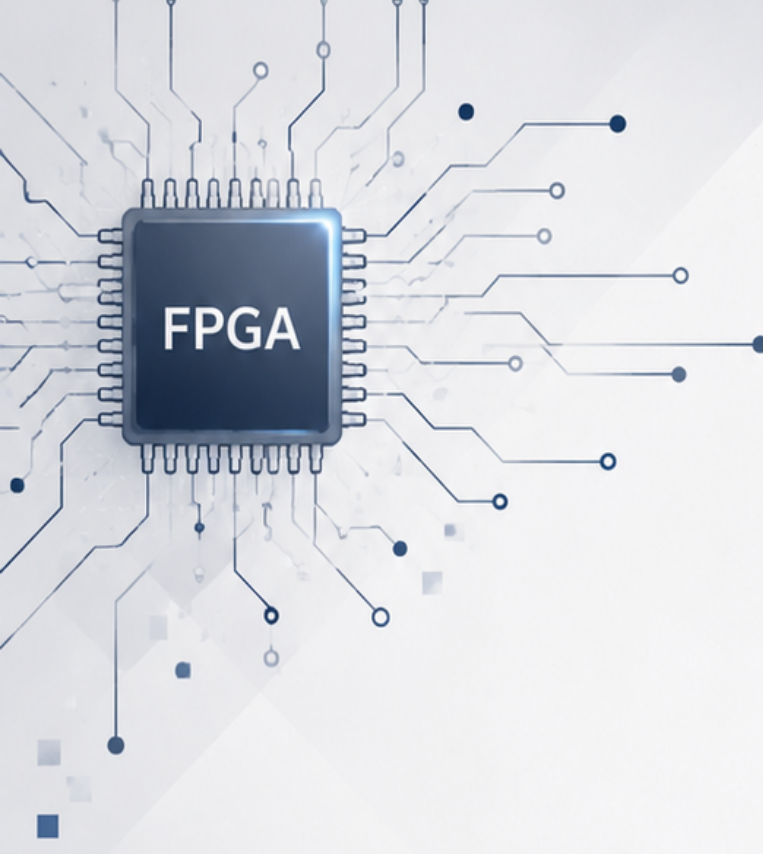




FPGA

Spartan6 - DSP48A1

Digital Design Diploma

Mayar Walid Gamal

Email: mayar1.walid@gmail.com



Contents

1	RTL Code	2
1.1	Sub Modules	2
1.1.1	Register Module	2
1.1.2	Parameterized Multiplexer Module	3
1.1.3	Pipeline Stage Module	3
1.1.4	Operator Module	4
1.1.5	4x1 Multiplexer Module	4
1.1.6	2x1 Multiplexer Module	5
1.2	Top Module	5
1.2.1	DSP48A1 Module	5
2	Testbench Code	8
2.1	DSP48A1_tb Module	8
3	Do file	12
4	QuestaSim Snippets	13
5	Constraint File	14
6	Elaboration	15
6.1	Messages Tab	15
6.2	Schematic	15
7	Synthesis	16
7.1	Messages Tab	16
7.2	Utilization Report	16
7.3	Timing Report	17
7.4	Schematic	17
8	Implementation	18
8.1	Messages Tab	18
8.2	Utilization Report	18
8.3	Timing Report	19
8.4	Device	19
9	Linting	20

1 RTL Code

1.1 Sub Modules

1.1.1 Register Module

```
1  module REG (d,clk,rst,en,q);
2      parameter N = 1;
3      parameter RSTTYPE_REG = "SYNC";
4      input [N-1:0] d;
5      input clk,rst,en;
6      output reg [N-1:0] q;
7
8      generate
9          if (RSTTYPE_REG == "ASYNC") begin
10             always @(posedge clk or posedge rst) begin
11                 if (rst) begin
12                     q <= 0;
13                 end
14                 else if (en) begin
15                     q <= d;
16                 end
17             end
18         end
19         else if (RSTTYPE_REG == "SYNC") begin
20             always @(posedge clk) begin
21                 if (rst) begin
22                     q <= 0;
23                 end
24                 else begin
25                     q <= d;
26                 end
27             end
28         end
29     endgenerate
30 endmodule
```

1.1.2 Parameterized Multiplexer Module

```
1  module ParamMux #(
2      parameter REGISTER = 1,
3      parameter N = 1
4  )(
5      input [N-1:0] x,y,
6      output reg [N-1:0] out
7  );
8      always @(*) begin
9          if (REGISTER == 1) begin
10             out = x;
11         end
12         else begin
13             out = y;
14         end
15     end
16 endmodule
```

1.1.3 Pipeline Stage Module

```
1  module REG_BLK #(
2      parameter N = 1,
3      parameter REGISTER = 1,
4      parameter RSTTYPE = "SYNC"
5  )(
6      input [N-1:0] in,
7      input clk,rst,en,
8      output [N-1:0] out
9  );
10     wire [N-1:0] z;
11
12     REG #(.N(N),.RSTTYPE_REG(RSTTYPE)) REGBLK
13     ↪ (.d(in),.clk(clk),.rst(rst),.en(en),.q(z));
14     ParamMux #(.N(N),.REGISTER(REGISTER)) MUXBLK (.x(z),.y(in),.out(out));
15 endmodule
```

1.1.4 Operator Module

```
1  module OPERATOR #(
2      parameter N = 1
3  )(
4      input [N-1:0] x,y,
5      input opmode,cin,
6      output reg cout,
7      output reg [N-1:0] out
8  );
9      reg [N:0] temp;
10     always @(*) begin
11         if (opmode == 0) begin
12             temp = x + y + cin;
13         end
14         else begin
15             temp = x - ( y + cin ) ;
16         end
17
18         out = temp[N-1:0];
19         cout = temp[N];
20     end
21 endmodule
```

1.1.5 4x1 Multiplexer Module

```
1  module MUX41 #(
2      parameter N = 1
3  )(
4      input [1:0] opmode,
5      input [N-1:0] x,y,z,
6      output [N-1:0] out
7  );
8      assign out = (opmode == 2'b00)? {N{1'b0}} :
9                  (opmode == 2'b01)? x :
10                 (opmode == 2'b10)? y : z ;
11 endmodule
```

1.1.6 2x1 Multiplexer Module

```
1  module MUX21 #(
2      parameter N = 1
3  )(
4      input opmode,
5      input [N-1:0] x,y,
6      output [N-1:0] out
7  );
8      assign out = (opmode == 1'b0)? x : y ;
9  endmodule
```

1.2 Top Module

1.2.1 DSP48A1 Module

```
1  module DSP48A1 #(
2      parameter AOREG = 0,
3      parameter A1REG = 1,
4      parameter BOREG = 0,
5      parameter B1REG = 1,
6      parameter CREG = 1,
7      parameter DREG = 1,
8      parameter MREG = 1,
9      parameter PREG = 1,
10     parameter CARRYINREG = 1,
11     parameter CARRYOUTREG = 1,
12     parameter OPMODEREG = 1,
13     parameter CARRYINSEL = "OPMODE5",
14     parameter B_INPUT = "DIRECT",
15     parameter RSTTYPE = "SYNC"
16 )(
17     input [17:0] A,B,D,BCIN,
18     input [47:0] C,PCIN,
19     input [7:0] OPMODE,
20     input CARRYIN,CLK,
21     input CEA,CEB,CEC,CECARRYIN,CED,CEM,CEOPMODE,CEP,
22     input RSTA,RSTB,RSTC,RSTD,RSTCARRYIN,RSTM,RSTOPMODE,RSTP,
23     output [35:0] M,
24     output [47:0] P,PCOUT,
25     output CARRYOUT,CARRYOUTF,
26     output [17:0] BCOUT
27 );
28     wire [17:0] AO_REG,BO_REG,D_REG,PREOUT,B1,A1_REG;
29     wire [47:0] C_REG,DAB,X,Z,POSTOUT;
30     wire [35:0] M_OUT,M_REG;
31     wire [7:0] OPMODE_REG;
32     wire CIN,COOUT;
```

```

33
34 REG_BLK #(.N(8),.REGISTER(OPMODEREG),.RSTTYPE(RSTTYPE)) OPREG
    ↪ (.in(OPMODE),.clk(CLK),.out(OPMODE_REG),.rst(RSTOPMODE),.en(CEOPMODE));
35
36 REG_BLK #(.N(18),.REGISTER(AOREG),.RSTTYPE(RSTTYPE)) AO
    ↪ (.in(A),.clk(CLK),.out(AO_REG),.rst(RSTA),.en(CEA));
37
38 generate
39     if (B_INPUT == "DIRECT") begin
40         REG_BLK #(.N(18),.REGISTER(BOREG),.RSTTYPE(RSTTYPE)) BO
41             ↪ (.in(B),.clk(CLK),.out(BO_REG),.rst(RSTB),.en(CEB));
42     end
43     else if (B_INPUT == "CASCADE") begin
44         REG_BLK #(.N(18),.REGISTER(BOREG),.RSTTYPE(RSTTYPE)) BO
45             ↪ (.in(BCIN),.clk(CLK),.out(BO_REG),.rst(RSTB),.en(CEB));
46     end
47     else begin
48         REG_BLK #(.N(18),.REGISTER(BOREG),.RSTTYPE(RSTTYPE)) BO
49             ↪ (.in(BCIN),.clk(CLK),.out(1'b0),.rst(RSTB),.en(CEB));
50     end
51 endgenerate
52
53 REG_BLK #(.N(48),.REGISTER(CREG),.RSTTYPE(RSTTYPE)) CO
54     ↪ (.in(C),.clk(CLK),.out(C_REG),.rst(RSTC),.en(CEC));
55 REG_BLK #(.N(18),.REGISTER(DREG),.RSTTYPE(RSTTYPE)) DO
56     ↪ (.in(D),.clk(CLK),.out(D_REG),.rst(RSTD),.en(CED));
57
58 OPERATOR #(.N(18)) PreOperator
59     ↪ (.x(D_REG),.y(BO_REG),.opmode(OPMODE_REG[6]),.out(PREOUT),.cin(1'b0),.cout());
60 MUX21 #(.N(18)) MUX (.x(BO_REG),.y(PREOUT),.out(B1),.opmode(OPMODE_REG[4]));
61
62 REG_BLK #(.N(18),.REGISTER(B1REG),.RSTTYPE(RSTTYPE)) B1_REG
63     ↪ (.in(B1),.clk(CLK),.out(BCOUT),.rst(RSTB),.en(CEB));
64 REG_BLK #(.N(18),.REGISTER(A1REG),.RSTTYPE(RSTTYPE)) A1
65     ↪ (.in(AO_REG),.clk(CLK),.out(A1_REG),.rst(RSTA),.en(CEA));
66 assign M_OUT = BCOUT * A1_REG;
67
68 REG_BLK #(.N(36),.REGISTER(MREG),.RSTTYPE(RSTTYPE)) MO
69     ↪ (.in(M_OUT),.clk(CLK),.out(M_REG),.rst(RSTM),.en(CEM));
70 assign M = M_REG;
71
72 generate
73     if (CARRYINSEL == "CARRYIN") begin
74         REG_BLK #(.N(1),.REGISTER(CARRYINREG),.RSTTYPE(RSTTYPE)) CYI
75             ↪ (.in(CARRYIN),.clk(CLK),.out(CIN),.rst(RSTCARRYIN),.en(CECARRYIN));
76     end
77     else if (CARRYINSEL == "OPMODE5") begin
78         REG_BLK #(.N(1),.REGISTER(CARRYINREG),.RSTTYPE(RSTTYPE)) CYI
79             ↪ (.in(OPMODE_REG[5]),.clk(CLK),.out(CIN),.rst(RSTCARRYIN),
80                 .en(CECARRYIN));
81     end
82     else begin

```

```

72     REG_BLK #(.N(1),.REGISTER(CARRYINREG),.RSTTYPE(RSTTYPE)) CYI
       ↳ (.in(OPMODE_REG[5]),.clk(CLK),.out(1'b0),.rst(RSTCARRYIN),
73         .en(CECARRYIN));
74     end
75 endgenerate
76
77 assign DAB = {D_REG[11:0],A1_REG,BCOUT};
78
79 MUX41 #(.N(48)) MUX_X
       ↳ (.x({12'b0,M_REG}),.y(P),.z(DAB),.opmode(OPMODE_REG[1:0]),.out(X));
80 MUX41 #(.N(48)) MUX_Z
       ↳ (.x(PCIN),.y(P),.z(C_REG),.opmode(OPMODE_REG[3:2]),.out(Z));
81
82 OPERATOR #(.N(48)) PostOperator
       ↳ (.x(Z),.y(X),.opmode(OPMODE_REG[7]),.out(POSTOUT),.cin(CIN),.cout(COUT));
83
84 REG_BLK #(.N(1),.REGISTER(CARRYOUTREG),.RSTTYPE(RSTTYPE)) CYO
       ↳ (.in(COUT),.clk(CLK),.out(CARRYOUT),.rst(RSTCARRYIN),.en(CECARRYIN));
85 assign CARRYOUTF = CARRYOUT;
86
87 REG_BLK #(.N(48),.REGISTER(PREG),.RSTTYPE(RSTTYPE)) P_REG
       ↳ (.in(POSTOUT),.clk(CLK),.out(P),.rst(RSTP),.en(CEP));
88 assign PCOUT = P;
89
90 endmodule

```


2 Testbench Code

2.1 DSP48A1_tb Module

```
1  module DSP48A1_tb ();
2      reg[17:0] A_tb,B_tb,D_tb,BCIN_tb;
3      reg[47:0] C_tb,PCIN_tb;
4      reg[7:0] OPMODE_tb;
5      reg CARRYIN_tb,CLK_tb;
6      reg CEA_tb,CEB_tb,CEC_tb,CECARRYIN_tb,CED_tb,CEM_tb,CEOPMODE_tb,CEP_tb;
7      reg RSTA_tb,RSTB_tb,RSTC_tb,RSTD_tb,RSTCARRYIN_tb,RSTM_tb,RSTOPMODE_tb
8          ,RSTP_tb;
9      reg [35:0] M_exp;
10     wire [35:0] M_tb;
11     wire [47:0] P_dut;
12     reg [47:0] P_exp;
13     wire [47:0] PCOUT_tb;
14     wire CARRYOUT_tb,CARRYOUTF_tb;
15     reg CARRYOUT_exp;
16     wire [17:0] BCOUT_tb;
17     reg [17:0] BCOUT_exp;
18
19     DSP48A1
20         DUT (.A(A_tb),.B(B_tb),.C(C_tb),.D(D_tb),.CLK(CLK_tb)
21             ,.CARRYIN(CARRYIN_tb),.OPMODE(OPMODE_tb),.BCIN(BCIN_tb)
22             ,.PCIN(PCIN_tb),.RSTA(RSTA_tb),.RSTB(RSTB_tb),.RSTC(RSTC_tb)
23             ,.RSTD(RSTD_tb),.RSTM(RSTM_tb),.RSTP(RSTP_tb)
24             ,.RSTCARRYIN(RSTCARRYIN_tb),.RSTOPMODE(RSTOPMODE_tb)
25             ,.CEA(CEA_tb),.CEB(CEB_tb),.CEC(CEC_tb),.CED(CED_tb)
26             ,.CEM(CEM_tb),.CEP(CEP_tb),.CECARRYIN(CECARRYIN_tb)
27             ,.CEOPMODE(CEOPMODE_tb),.BCOUT(BCOUT_tb),.PCOUT(PCOUT_tb)
28             ,.P(P_dut),.M(M_tb),.CARRYOUT(CARRYOUT_tb)
29             ,.CARRYOUTF(CARRYOUTF_tb));
30
31     initial begin
32         CLK_tb = 0;
33         forever begin
34             #1 CLK_tb = ~CLK_tb;
35         end
36     end
37
38     initial begin
39         RSTA_tb = 1; RSTB_tb = 1; RSTC_tb = 1; RSTD_tb = 1;
40         RSTM_tb = 1; RSTP_tb = 1; RSTCARRYIN_tb = 1; RSTOPMODE_tb = 1;
41         //////////////////////////////////////////
42         A_tb = 18'd13; B_tb = 18'd25; C_tb = 48'd5; D_tb = 18'd59;
43         PCIN_tb = 48'd25; BCIN_tb = 18'd17; CARRYIN_tb = 1;
44         OPMODE_tb = 8'b0000_0001;
45         @(negedge CLK_tb);
46
47         if (P_dut != 48'd0)begin
48             $display("P mismatch: P = %0d", P_dut);
```

```

49         $stop;
50     end
51     if (M_tb != 36'd0)begin
52         $display("M mismatch: M = %0d", M_tb);
53         $stop;
54     end
55     if (BCOUT_tb != 18'd0)begin
56         $display("BCOUT mismatch: BCOUT = %0d", BCOUT_tb);
57         $stop;
58     end
59     if (PCOUT_tb != 48'd0)begin
60         $display("PCOUT mismatch: PCOUT = %0d", PCOUT_tb);
61         $stop;
62     end
63     if (CARRYOUT_tb != 1'b0)begin
64         $display("CARRYOUT mismatch: CARRYOUT = %0d", CARRYOUT_tb);
65         $stop;
66     end
67     if (CARRYOUTF_tb != 1'b0)begin
68         $display("CARRYOUTF mismatch: CARRYOUTF = %0d", CARRYOUTF_tb);
69         $stop;
70     end
71     ////////////////////////////////////////////Path 1
72     RSTA_tb = 0; RSTB_tb = 0; RSTC_tb = 0; RSTD_tb = 0;
73     RSTM_tb = 0; RSTP_tb = 0; RSTCARRYIN_tb = 0; RSTOPMODE_tb = 0;
74
75     CEA_tb = 1; CEB_tb = 1; CEC_tb = 1; CED_tb = 1;
76     CEM_tb = 1; CEP_tb = 1; CECARRYIN_tb = 1; CEOPMODE_tb = 1;
77
78     A_tb = 18'd20; B_tb = 18'd10; C_tb = 48'd350; D_tb = 18'd25;
79     PCIN_tb = 48'd5; BCIN_tb = 18'd15; CARRYIN_tb = 0; OPMODE_tb =
80     ↪ 8'b1101_1101;
81     repeat(4)@(negedge CLK_tb);
82     P_exp = 48'h32; M_exp = 36'h12c; BCOUT_exp = 18'hf; CARRYOUT_exp = 0;
83
84     if (P_dut != P_exp)begin
85         $display("P mismatch: P = %0d", P_dut);
86         $stop;
87     end
88     if (M_tb != M_exp)begin
89         $display("M mismatch: M = %0d", M_tb);
90         $stop;
91     end
92     if (BCOUT_tb != BCOUT_exp)begin
93         $display("BCOUT mismatch: BCOUT = %0d", BCOUT_tb);
94         $stop;
95     end
96     if (PCOUT_tb != P_exp)begin
97         $display("PCOUT mismatch: PCOUT = %0d", PCOUT_tb);
98         $stop;
99     end
100     if (CARRYOUT_tb != CARRYOUT_exp)begin

```

```

100         $display("CARRYOUT mismatch: CARRYOUT = %0d", CARRYOUT_tb);
101         $stop;
102     end
103     if (CARRYOUTF_tb != CARRYOUT_exp)begin
104         $display("CARRYOUTF mismatch: CARRYOUTF = %0d", CARRYOUTF_tb);
105         $stop;
106     end
107     $display("Path 1 passed");
108     ////////////////////////////////////////////Path 2
109     A_tb = 18'd20; B_tb = 18'd10; C_tb = 48'd350; D_tb = 18'd25;
110     PCIN_tb = 48'd5; BCIN_tb = 18'd15; CARRYIN_tb = 0; OPMODE_tb =
111     ↪ 8'b0001_0000;
112     repeat(3)@(negedge CLK_tb);
113     P_exp = 48'h0; M_exp = 36'h2bc; BCOUT_exp = 18'h23; CARRYOUT_exp = 0;
114
115     if (P_dut != P_exp)begin
116         $display("P mismatch: P = %0d", P_dut);
117         $stop;
118     end
119     if (M_tb != M_exp)begin
120         $display("M mismatch: M = %0d", M_tb);
121         $stop;
122     end
123     if (BCOUT_tb != BCOUT_exp)begin
124         $display("BCOUT mismatch: BCOUT = %0d", BCOUT_tb);
125         $stop;
126     end
127     if (PCOUT_tb != P_exp)begin
128         $display("PCOUT mismatch: PCOUT = %0d", PCOUT_tb);
129         $stop;
130     end
131     if (CARRYOUT_tb != CARRYOUT_exp)begin
132         $display("CARRYOUT mismatch: CARRYOUT = %0d", CARRYOUT_tb);
133         $stop;
134     end
135     if (CARRYOUTF_tb != CARRYOUT_exp)begin
136         $display("CARRYOUTF mismatch: CARRYOUTF = %0d", CARRYOUTF_tb);
137         $stop;
138     end
139     $display("Path 2 passed");
140     ////////////////////////////////////////////Path 3
141     A_tb = 18'd20; B_tb = 18'd10; C_tb = 48'd350; D_tb = 18'd25;
142     PCIN_tb = 48'd5; BCIN_tb = 18'd15; CARRYIN_tb = 0; OPMODE_tb =
143     ↪ 8'b0000_1010;
144     repeat(3)@(negedge CLK_tb);
145     M_exp = 36'hc8; BCOUT_exp = 18'ha;
146
147     if (P_dut != P_exp)begin
148         $display("P mismatch: P = %0d", P_dut);
149         $stop;
150     end
151     if (M_tb != M_exp)begin

```

```

150         $display("M mismatch: M = %0d", M_tb);
151         $stop;
152     end
153     if (BCOUT_tb != BCOUT_exp)begin
154         $display("BCOUT mismatch: BCOUT = %0d", BCOUT_tb);
155         $stop;
156     end
157     if (PCOUT_tb != P_exp)begin
158         $display("PCOUT mismatch: PCOUT = %0d", PCOUT_tb);
159         $stop;
160     end
161     if (CARRYOUT_tb != CARRYOUT_exp)begin
162         $display("CARRYOUT mismatch: CARRYOUT = %0d", CARRYOUT_tb);
163         $stop;
164     end
165     if (CARRYOUTF_tb != CARRYOUT_exp)begin
166         $display("CARRYOUTF mismatch: CARRYOUTF = %0d", CARRYOUTF_tb);
167         $stop;
168     end
169     $display("Path 3 passed");
170     ////////////////////////////////////////////Path 4
171     A_tb = 18'd5; B_tb = 18'd6; C_tb = 48'd350; D_tb = 18'd25;
172     PCIN_tb = 48'd3000; BCIN_tb = 18'd15; CARRYIN_tb = 0; OPMODE_tb =
173     ↪ 8'b1010_0111;
174     repeat(3)@(negedge CLK_tb);
175     P_exp = 48'hfe6ffec0bb1; M_exp = 36'h1e; BCOUT_exp = 18'h6;
176     ↪ CARRYOUT_exp = 1;
177
178     if (P_dut != P_exp)begin
179         $display("P mismatch: P = %0d", P_dut);
180         $stop;
181     end
182     if (M_tb != M_exp)begin
183         $display("M mismatch: M = %0d", M_tb);
184         $stop;
185     end
186     if (BCOUT_tb != BCOUT_exp)begin
187         $display("BCOUT mismatch: BCOUT = %0d", BCOUT_tb);
188         $stop;
189     end
190     if (PCOUT_tb != P_exp)begin
191         $display("PCOUT mismatch: PCOUT = %0d", PCOUT_tb);
192         $stop;
193     end
194     if (CARRYOUT_tb != CARRYOUT_exp)begin
195         $display("CARRYOUT mismatch: CARRYOUT = %0d", CARRYOUT_tb);
196         $stop;
197     end
198     if (CARRYOUTF_tb != CARRYOUT_exp)begin
199         $display("CARRYOUTF mismatch: CARRYOUTF = %0d", CARRYOUTF_tb);
200         $stop;
201     end

```

```

200         $display("All directed tests PASSED.");
201         $stop;
202     end
203
204     initial begin
205         $monitor("T=%0t | Expected P=%0d , Actual P=%0d , PCOUT=%0d| M=%0d , Actual
        ↳ M=%0d | BCOUT=%0d , Actual BCOUT=%0d| CARRYOUT=%b , Actual COUT=%0d |
        ↳ CARRYOUTF=%d",
206             $time, P_exp, P_dut,PCOUT_tb, M_exp,M_tb, BCOUT_exp,BCOUT_tb,
        ↳ CARRYOUT_exp,CARRYOUT_tb,CARRYOUTF_tb);
207     end
208 endmodule

```

3 Do file

```

1  vlib work
2  vlog DSP48A1.v DSP48A1_tb.v MUX41.v MUX21.v OPERATOR.v ParamMux.v REG.v
   ↳ REG_BLK.v
3  vsim -voptargs=+acc work.DSP48A1_tb
4  add wave *
5  run -all
6  #quit -sim

```

4 QuestaSim Snippets

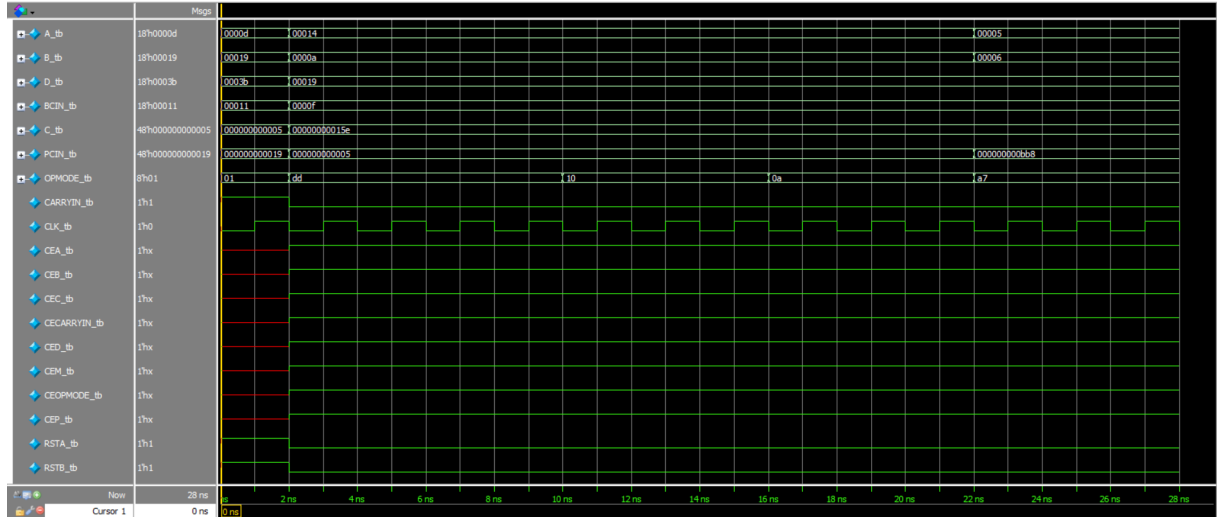


Figure 1: Waveform

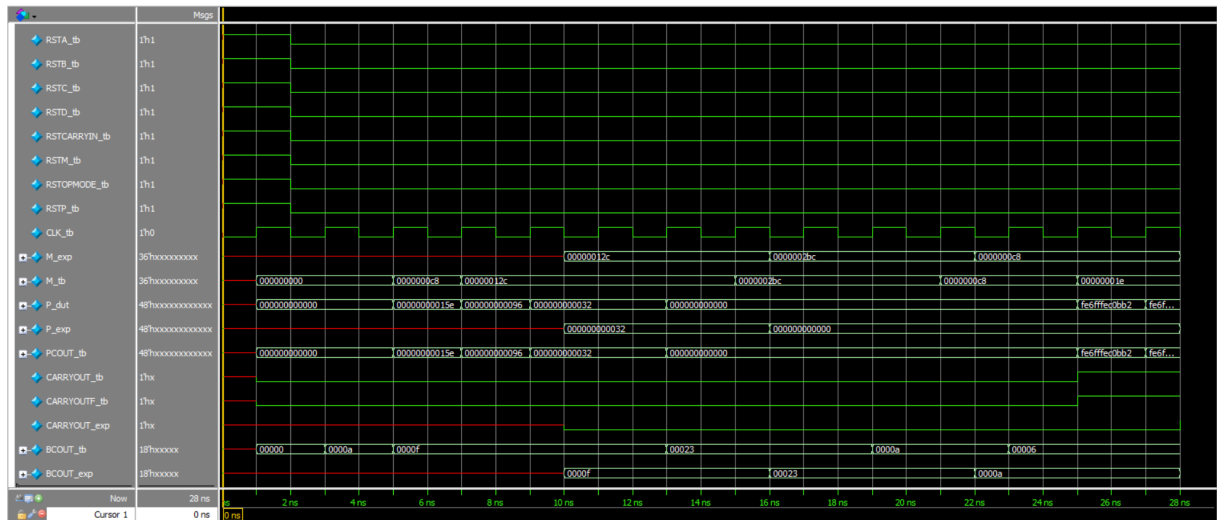


Figure 2: Waveform

```

# T=0 | Expected P=x , Actual P=x , PCOUT=x| M=x , Actual M=x | BCOUT=x , Actual BCOUT=x| CARRYOUT=x , Actual COUT=x | CARRYOUTF=x
# T=1 | Expected P=x , Actual P=0 , PCOUT=0| M=x , Actual M=0 | BCOUT=x , Actual BCOUT=0| CARRYOUT=x , Actual COUT=0 | CARRYOUTF=0
# T=2 | Expected P=50 , Actual P=0 , PCOUT=0| M=300 , Actual M=0 | BCOUT=15 , Actual BCOUT=0| CARRYOUT=0 , Actual COUT=0 | CARRYOUTF=0
# T=3 | Expected P=50 , Actual P=0 , PCOUT=0| M=300 , Actual M=0 | BCOUT=15 , Actual BCOUT=10| CARRYOUT=0 , Actual COUT=0 | CARRYOUTF=0
# T=5 | Expected P=50 , Actual P=350 , PCOUT=350| M=300 , Actual M=200 | BCOUT=15 , Actual BCOUT=15| CARRYOUT=0 , Actual COUT=0 | CARRYOUTF=0
# T=7 | Expected P=50 , Actual P=150 , PCOUT=150| M=300 , Actual M=300 | BCOUT=15 , Actual BCOUT=15| CARRYOUT=0 , Actual COUT=0 | CARRYOUTF=0
# T=9 | Expected P=50 , Actual P=50 , PCOUT=50| M=300 , Actual M=300 | BCOUT=15 , Actual BCOUT=15| CARRYOUT=0 , Actual COUT=0 | CARRYOUTF=0
# Path 1 passed
# T=10 | Expected P=0 , Actual P=50 , PCOUT=50| M=700 , Actual M=300 | BCOUT=35 , Actual BCOUT=15| CARRYOUT=0 , Actual COUT=0 | CARRYOUTF=0
# T=13 | Expected P=0 , Actual P=0 , PCOUT=0| M=700 , Actual M=300 | BCOUT=35 , Actual BCOUT=35| CARRYOUT=0 , Actual COUT=0 | CARRYOUTF=0
# T=15 | Expected P=0 , Actual P=0 , PCOUT=0| M=700 , Actual M=300 | BCOUT=35 , Actual BCOUT=35| CARRYOUT=0 , Actual COUT=0 | CARRYOUTF=0
# Path 2 passed
# T=16 | Expected P=0 , Actual P=0 , PCOUT=0| M=200 , Actual M=700 | BCOUT=10 , Actual BCOUT=35| CARRYOUT=0 , Actual COUT=0 | CARRYOUTF=0
# T=19 | Expected P=0 , Actual P=0 , PCOUT=0| M=200 , Actual M=700 | BCOUT=10 , Actual BCOUT=10| CARRYOUT=0 , Actual COUT=0 | CARRYOUTF=0
# T=21 | Expected P=0 , Actual P=0 , PCOUT=0| M=200 , Actual M=200 | BCOUT=10 , Actual BCOUT=10| CARRYOUT=0 , Actual COUT=0 | CARRYOUTF=0
# Path 3 passed
# T=22 | Expected P=279756988484529 , Actual P=0 , PCOUT=0| M=30 , Actual M=200 | BCOUT=6 , Actual BCOUT=10| CARRYOUT=1 , Actual COUT=0 | CARRYOUTF=0
# T=23 | Expected P=279756988484529 , Actual P=0 , PCOUT=0| M=30 , Actual M=200 | BCOUT=6 , Actual BCOUT=6| CARRYOUT=1 , Actual COUT=0 | CARRYOUTF=0
# T=25 | Expected P=279756988484529 , Actual P=279756988484530 , PCOUT=279756988484530| M=30 , Actual M=30 | BCOUT=6 , Actual BCOUT=6| CARRYOUT=1 , Actual COUT=1 | CARRYOUTF=1
# T=27 | Expected P=279756988484529 , Actual P=279756988484529 , PCOUT=279756988484529| M=30 , Actual M=30 | BCOUT=6 , Actual BCOUT=6| CARRYOUT=1 , Actual COUT=1 | CARRYOUTF=1
# All directed tests PASSED.
# ** Note: $stop : DSP48A1_tb.v(198)
# Time: 28 ns Iteration: 1 Instance: /DSP48A1_tb
# Break in Module DSP48A1_tb at DSP48A1_tb.v line 198

```

Figure 3: Transcript

5 Constraint File

```

1  ## Clock signal
2  set_property -dict { PACKAGE_PIN W5    IOSTANDARD LVCMOS33 } [get_ports CLK]
3  create_clock -add -name sys_clk_pin -period 10.000 -waveform {0 5} [get_ports
4  ↪ CLK]

```

6 Elaboration

6.1 Messages Tab

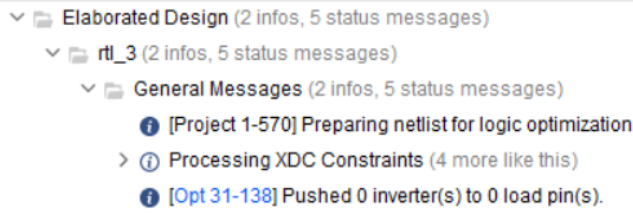


Figure 4: Messages tab

6.2 Schematic

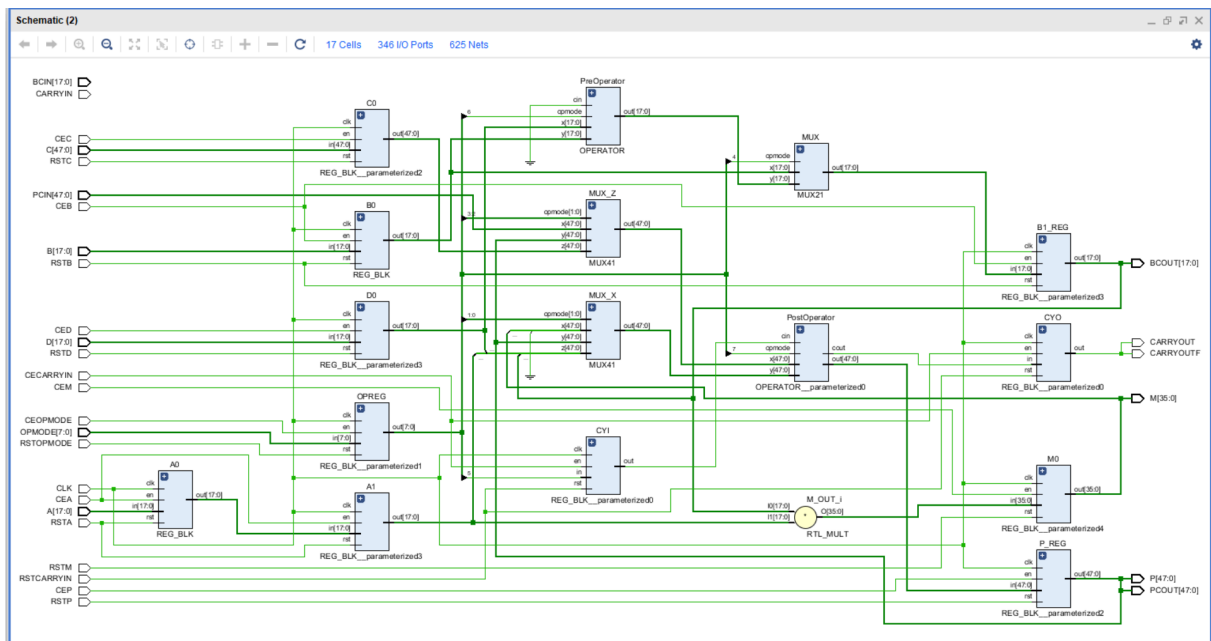


Figure 5: Schematic

7 Synthesis

7.1 Messages Tab

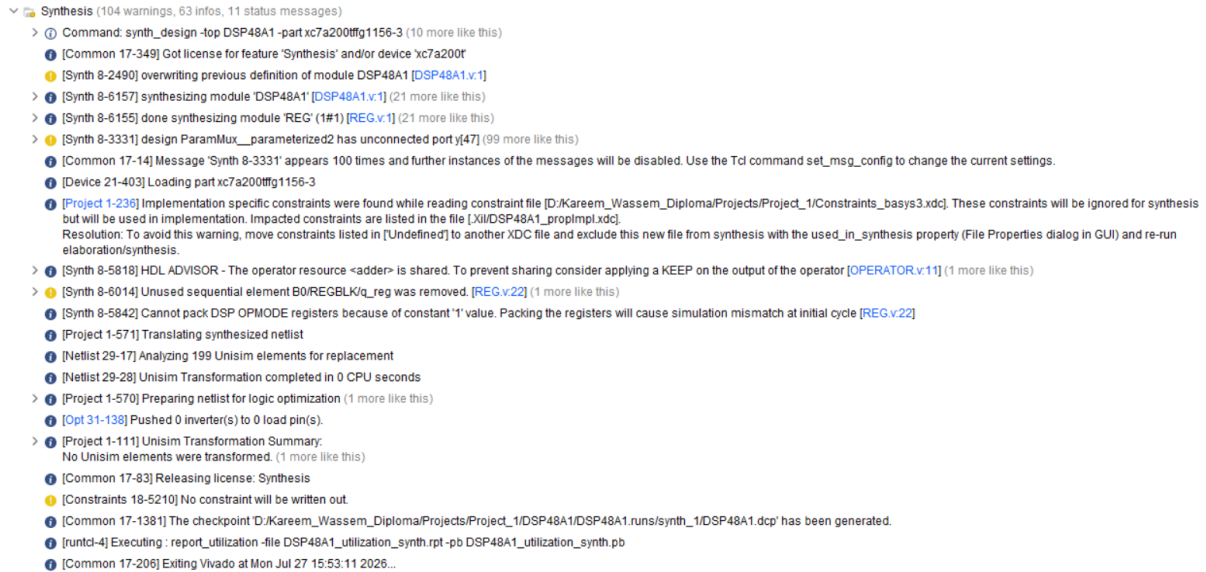


Figure 6: Messages tab

7.2 Utilization Report

Tcl Console	Messages	Log	Reports	Design Runs	Utilization	Timing	Debug
Hierarchy							
Hierarchy							
Summary							
Slice Logic							
Slice LUTs (<1%)							
LUT as Logic (<1%)							
Slice Registers (<1%)							
Register as Flip Flop (<1%)							
Memory							
DSP							
DSPs (<1%)							
DSP48E1 only							
IO and GT Specific							
Bonded IOB (64%)							
IOB Master Pads							
Clocking							
BUFGCTRL (3%)							
Specific Feature							
Primitives							
Black Boxes							
Instantiated Netlists							
Name	Slice LUTs (134600)	Slice Registers (269200)	DSPs (740)	Bonded IOB (500)	BUFGCTRL (32)		
DSP48A1	230	160	1	319	1		
A1 (REG_BLK__para...	0	18	0	0	0		
B1_REG (REG_BLK__...	0	18	0	0	0		
C0 (REG_BLK__para...	0	48	0	0	0		
CYI (REG_BLK__para...	0	1	0	0	0		
CYO (REG_BLK__par...	0	1	0	0	0		
D0 (REG_BLK__para...	36	18	0	0	0		
M0 (REG_BLK__para...	0	0	1	0	0		
OPREG (REG_BLK__...	194	8	0	0	0		
P_REG (REG_BLK__p...	0	48	0	0	0		
PostOperator (OPERA...	0	0	0	0	0		
PreOperator (OPERAT...	0	0	0	0	0		

Figure 7: Utilization

7.3 Timing Report

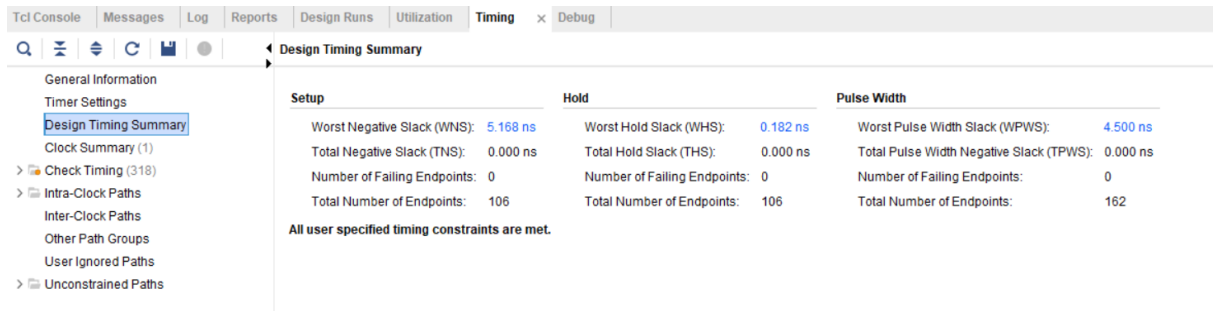


Figure 8: Timing

7.4 Schematic

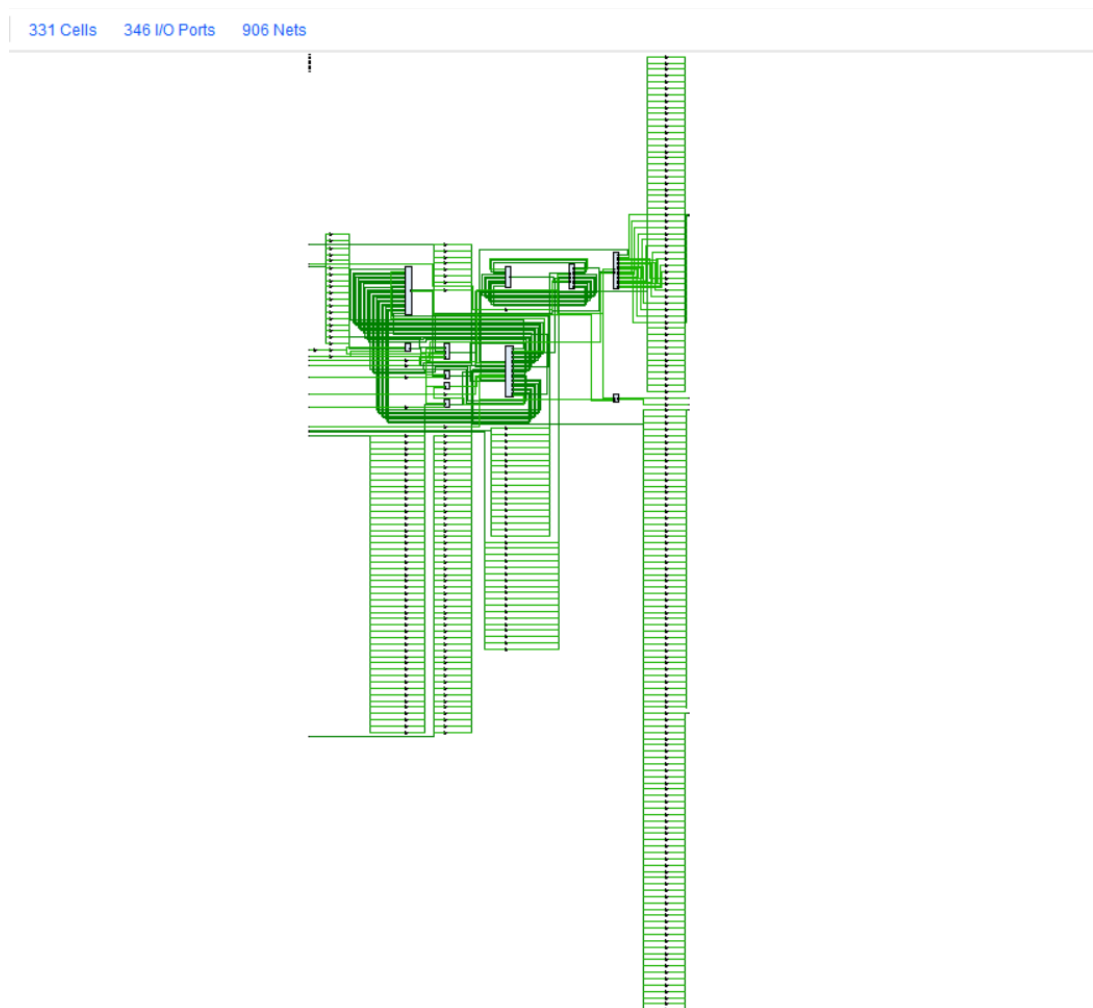


Figure 9: Schematic

8 Implementation

8.1 Messages Tab

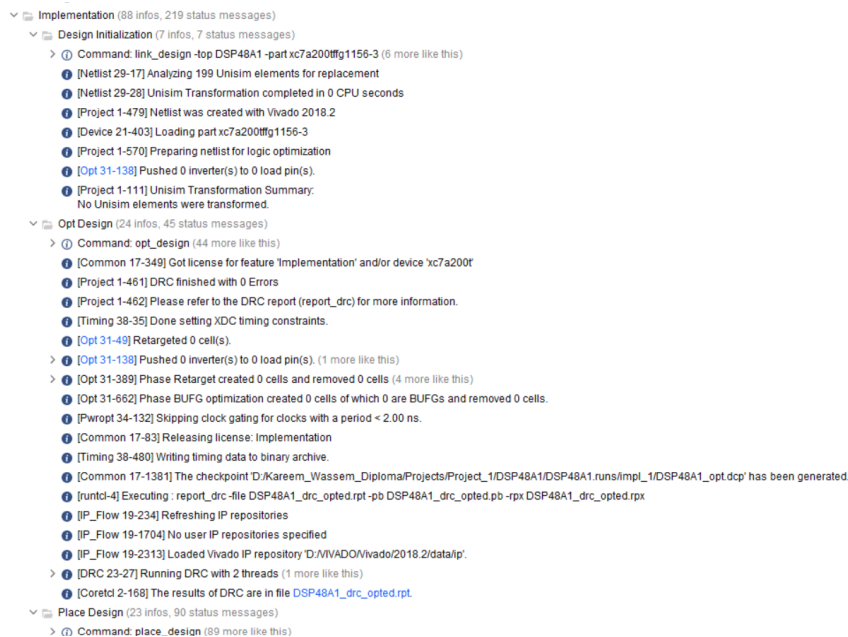


Figure 10: Messages tab

8.2 Utilization Report

Tcl Console Messages Log Reports Design Runs Power DRC Methodology Timing Utilization									
Hierarchy									
Name	Slice LUTs (133800)	Slice Registers (267600)	Slice (33450)	LUT as Logic (133800)	LUT Flip Flop Pairs (133800)	DSP s (740)	Bonded IOB (500)	BUFCTRL (32)	
DSP48A1	230	179	110	230	50	1	319	1	
> A1 (REG_BLK__para...	0	18	9	0	0	0	0	0	
> B1_REG (REG_BLK__...	0	36	19	0	0	0	0	0	
> C0 (REG_BLK__para...	0	48	17	0	0	0	0	0	
> C1 (REG_BLK__para...	0	1	1	0	0	0	0	0	
> C2 (REG_BLK__par...	0	2	2	0	0	0	0	0	
> D0 (REG_BLK__para...	36	18	15	36	0	0	0	0	
> M0 (REG_BLK__para...	0	0	0	0	0	1	0	0	
> OPREG (REG_BLK__...	194	8	63	194	0	0	0	0	
> P_REG (REG_BLK__p...	0	48	12	0	0	0	0	0	
PostOperator (OPERA...	0	0	13	0	0	0	0	0	
PreOperator (OPERA...	0	0	5	0	0	0	0	0	

Figure 11: Utilization

8.3 Timing Report

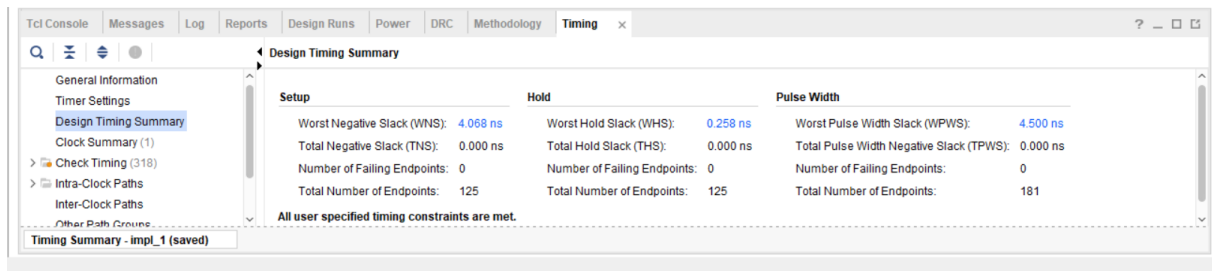


Figure 12: Timing

8.4 Device

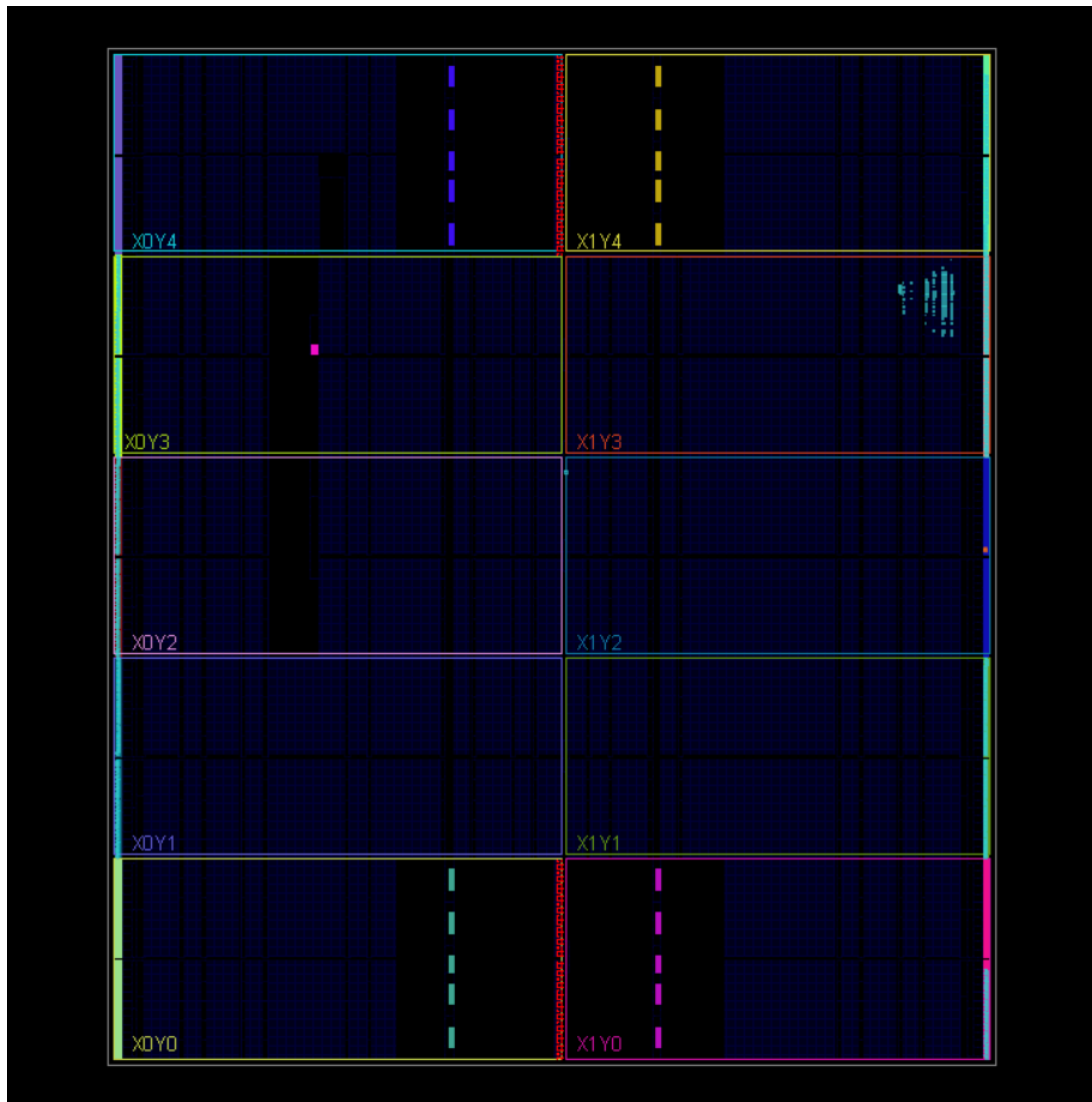


Figure 13: Device

9 Linting

The screenshot shows the Lint Checks window in a design tool. The main window displays the code for a module named DSP48A1. The Lint Checks window shows a list of linting errors. The Lint Summary on the right shows 13 errors, with 13 (20) Open/uninspected, pending and 13 (20) Info.

Severity	Status	Check	Alias	Message	Module	Category	State	Owner	STARC Reference
Info	Open	parameter_name_duplicate		Same parameter name is used in more than one mod...	MUX21	Nomenclature...	open	unassign...	1.1.4.3, 3.2.2.2
Info	Open	parameter_name_duplicate		Same parameter name is used in more than one mod...	ParamMux	Nomenclature...	open	unassign...	1.1.4.3, 3.2.2.2
Info	Open	parameter_name_duplicate		Same parameter name is used in more than one mod...	DSP48A1	Nomenclature...	open	unassign...	1.1.4.3, 3.2.2.2
Info	Open	line_char_large		Line has more characters than the specified limit. Curr...	none	Rtl Design Style	open	unassign...	3.1.4.5
Info	Open	line_char_large		Line has more characters than the specified limit. Curr...	none	Rtl Design Style	open	unassign...	3.1.4.5
Info	Open	multi_ports_in_single_line		Multiple ports are declared in one line. Module DSP48...	DSP48A1	Rtl Design Style	open	unassign...	3.5.6.3
Info	Open	multi_ports_in_single_line		Multiple ports are declared in one line. Module MUX2...	MUX21	Rtl Design Style	open	unassign...	3.5.6.3
Info	Open	multi_ports_in_single_line		Multiple ports are declared in one line. Module MUX4...	MUX41	Rtl Design Style	open	unassign...	3.5.6.3
Info	Open	multi_ports_in_single_line		Multiple ports are declared in one line. Module OPER...	OPERATOR	Rtl Design Style	open	unassign...	3.5.6.3
Info	Open	multi_ports_in_single_line		Multiple ports are declared in one line. Module Param...	ParamMux	Rtl Design Style	open	unassign...	3.5.6.3
Info	Open	multi_ports_in_single_line		Multiple ports are declared in one line. Module REG...	REG	Rtl Design Style	open	unassign...	3.5.6.3
Info	Open	multi_ports_in_single_line		Multiple ports are declared in one line. Module REG...	REG_BLK	Rtl Design Style	open	unassign...	3.5.6.3

Figure 14: Linting